Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 7

7	eg 3^3 or $\left(3^{-2}\right)\times 3^{-5}$ or $\frac{3^3}{\left(3^{10}\right)}$ or $\frac{\left(3^5\right)}{3^{12}}$ or $\frac{\left(3^{-2}\right)}{3^5}$ or $3^{-12}\left(\times 3^5\right)$oe or $-2+5-10$ oe or $-12+5$ oe or $3-10$ oe		2	M1	for a correct application of an index rule as a first step or a correct calculation for n
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	-7		A1	Allow 3^{-7}
					Total 2 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 7

7	(a)		1	1	B1 cao
	(b)		-6	1	B1 Allow 3^{-6}
					Total 2 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 8

8	eg $(8^{-2}\times 8^9 =) 8^{-2+9}$ or 8^7 or 2^{-6+27} or 2^{21} **or** $(8^{-2}\div 8^{10} =) 8^{-2-10}$ or 8^{-12} or $\frac{1}{8^{12}}$ or 2^{-6-30} or 2^{-36} or $\frac{1}{2^{36}}$ **or** $(8^9\div 8^{10} =) 8^{9-10}$ or 8^{-1} or $\frac{1}{8^{(1)}}$ or 2^{27-30} or 2^{-3} or $\frac{1}{2^3}$ **or** $(8^n\times 8^{10} =) 8^{n+10}$ or 2^{3n+30} **OR** $-2+9=n+10$ oe eg $-6+27=3n+30$ oe **OR** $-2+9-10$ oe eg $\frac{-6+27-30}{3}$ oe		2	M1	for one correct application of an index rule (must be seen in powers of 8 or correct conversion to powers of 2) this could be after an initial mistake – working will need to be clearly seen **OR** for forming a correct equation in the indices alone **OR** for a complete method for the value of n
	Correct answer scores full marks (unless from obvious incorrect working)	-3		A1	Accept 8^{-3} or $\left(2^3\right)^{-3}$
					Total 2 marks

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2012 · Question 8

8.	$y^{3+n-1}=y^6$ oe or $y^{3+n}=y^7$ oe or $3+n-1=6$ oe or $y^n=\frac{y^7}{y^3}$ or $y^n=\frac{y^6}{y^2}$ or $y^n=y^4$		2	M1	*SC* if M0, award B1 for an answer of y^4
		4		A1 cao	
					Total 2 marks

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 10

10	(a)		0	1	B1 condone 150^0
	(b)		-2	1	B1 condone 3^{-2}
					Total 2 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 12

12	$2^{-4x} = 2^5$ or $-4x = 5$ or $-\frac{4}{5}x = 1$ oe		2	M1
	Correct answer scores full marks (unless from obvious incorrect working)	$-\frac{5}{4}$		A1 oe allow eg $\frac{5}{-4}$
				Total 2 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 12

12	(a)		$\frac{9}{10}$	1	B1 oe
	(b)		-6	1	B1
					Total 2 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 1

Figures in inverted commas must come from a correct method previously seen unless otherwise stated.

Q	Working	Answer	Mark	Notes
1	$\frac{12}{7}(\times)\frac{35}{16}$ oe		3	M1 for both fractions written as improper fractions
	$\frac{12}{7}\times\frac{35}{16}=\frac{420}{112}$ oe eg $\frac{192}{112}\times\frac{245}{112}=\frac{47040}{12544}$ or $\frac{\cancel{12}^{3}}{7^{1}}\times\frac{\cancel{35}^{5}}{\cancel{16}^{4}}$ oe or eg $\frac{12}{7^{1}}\times\frac{\cancel{35}^{5}}{16}=\frac{60}{16}$ oe			M1 for multiplying the numerators and multiplying the denominators **or** cancelling the fractions fully **or** cancelling fractions partially and multiplying across
	$\frac{12}{7}\times\frac{35}{16}=\frac{420}{112}=\frac{15}{4}=3\frac{3}{4}$ oe or $\frac{12}{7}\times\frac{35}{16}=\frac{420}{112}=3\frac{84}{112}=3\frac{3}{4}$ oe or $\frac{\cancel{12}^{3}}{7^{1}}\times\frac{\cancel{35}^{5}}{\cancel{16}^{4}}=\frac{15}{4}=3\frac{3}{4}$ oe *Working required*	shown		A1 completion to given result. dep on M2 If a student shows clearly in their working that $3\frac{3}{4}=\frac{15}{4}$ they only need to show that the LHS comes to $\frac{15}{4}$
				Total 3 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 4

4	(a)		2^7	1	B1 cao
	(b)	$\frac{280}{35}$ **or** $\frac{280}{5\times7}$ **or** 8 **or** $280 = 8\times5\times7$ **or** 2^3 **or** fully correct factor tree or repeated division or 2, 2, 2, 5, 7 or $2\times2\times2\times5\times7$		2	M1
			3		A1 cao
					Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 12

12	$3^4=\frac{3^x}{9^{3x}}$ **or** $81=\frac{3^x}{(3^2)^{3x}}$	$9^2=\frac{3^x}{9^{3x}}$ **or** $81=\frac{(9^{0.5})^x}{9^{3x}}$			M1 replacing 81 with 3^4 **or** 9^{3x} with $(3^2)^{3x}$ (or 3^{6x}) **or** replacing 81 with 9^2 **or** 3^x with $(9^{0.5})^x$ (in an equation)
	eg $4+6x=x$ or $4=x-2(3x)$ oe	eg $2=0.5x-3x$ oe			M1 a correct equation using powers
			-0.8	3	A1 oe, dep on at least M1
					Total 3 marks

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Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 14

14	eg $3^{2n+3-4}\left(=3^3\times3^{1-2n}\right)$ **or** $3^{2n-1}\left(=3^3\times3^{1-2n}\right)$ **or** $\left(3^{2n+3}=\right)3^7\times3^{1-2n}$ **or** $\left(\frac{3^{2n+3}}{3^4}=\right)3^{4-2n}$ **or** $\left(3^{2n+3}=\right)3^3\times3^{5-2n}$ **or** $\frac{3^{2n}}{3^4}=3^{1-2n}$ (division by 3) *This is not an exhaustive list* **or** $2n+3-4$ (= ….) **or** (… =) $3+1-2n$ or $2n-1$ (=…) **or** (... =) $4-2n$ (no other options for this)		3	M1	For one rule of indices used to correctly combine two or more of the **given** expressions (do not need part in brackets) *(must include algebra)* **or** For one of the 4 expressions shown, providing it is clear that they apply to the LHS or to the RHS
	eg $3^{2n+3-4}=3^{3+1-2n}$ **or** $3^{2n+3}=3^{8-2n}$ **or** $3^{2n-1}=3^{4-2n}$ **or** $2n+3-4=3+1-2n$ oe eg $2n-1=4-2n$			M1	A correct single power of 3 on both sides or a correct equation in n without indices (some students may go straight to this and gain M2)
	Working required	$\frac{5}{4}$		A1	oe dep on M1
					Total 3 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2018 · Question 14

14	$2^7=4^{2x}\times2^x$ or $128=(2^2)^{2x}\times2^x$	1.4	3	M1	Replacing 128 by 2^7 or 4 by 2^2
	$7=2(2x)+x$			M1	
				A1	oe
					Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2021 · Question 1

to imply a correct method

Q	Working	Answer	Mark	Notes
1 (a)		9	1	B1 allow 3^9
(b)		21	1	B1 allow 5^{21}
(c)	$8+2-p=6$ oe eg $8+2=6+p$ or $7^{8+2-p}=7^6$ oe		2	M1 (or embedded eg $8+2=10$, $10-4=6$)
	Correct answer scores full marks (unless from obvious incorrect working)	4		A1 allow 7^4
				Total 4 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2012 · Question 3

3. (a)		2^9	1	B1	cao	
(b)		3^5	1	B1	cao	
(c)	$5^{n-4-6}=5^3$ oe or $5^{n-10}=5^3$ oe or $n-4-6=3$ oe or $n-10=3$ oe or $5^n=5^3\times5^{10}$ oe or $5^n=5^{3+10}$ or $5^n=5^{13}$		2	M1		*SC* If M0, award B1 for an answer of 5^{13}
		13		A1	cao	
						Total 4 marks

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2022 · Question 6*

6	(a)		1	1	B1
	(b)		6	1	B1
	(c)	$206 + m - 214 = -3$ oe or $\frac{7^{-3} \times 7^{214}}{7^{206}}$ or $\frac{7^{211}}{7^{206}}$ oe		2	M1 allow $7^{206+m-214} = 7^{-3}$ oe (must be in the form $7^x = 7^y$ where x and y are correct expressions)
			5		A1 accept 7^5
					Total 4 marks

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2018 · Question 10*

10	(a)	$\frac{2^3}{2^7}$ **or** $2^3 \times 2^{-7}$ **or** $\frac{1}{2^4}$ **or** $\frac{1}{16}$ **and** $16 = 2^4$		2	M1
			−4		A1 Accept 2^{-4}
	(b)	$13^{-24} \times 13^5$		2	M1 for 13^{-24} or for $k = -6 \times 4 + 5$
			−19		A1 Accept 13^{-19}
					Total 4 marks

CHALLENGE QUESTIONS

Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 11*

11	(a)	$2^{\frac{1}{2}} \times 2^4$ **or** eg $2 \times (2^4)^2 = (2^x)^2$ or $2^9 = 2^{2x}$		2	M1	for a correct expression in powers of 2 that is equivalent to 2^x eg $2^{\frac{1}{2}} \times 2^4$ **or** for showing $\sqrt{2} = 2^{\frac{1}{2}}$ **and** $16 = 2^4$ **or** for writing the equation in powers of 2 eg $2 \times (2^4)^2 = (2^x)^2$ or $2^9 = 2^{2x}$
		Working required	$\frac{9}{2}$		A1	**or** 4.5 **or** 4½ dependent on M1
	(b)	$\frac{11^{-30}}{11^4}$ **or** $-30 - 4 = n$ **or** $-30 = n + 4$ oe		2	M1	For $(11^{-6})^5$ written as 11^{-30} in the equation or $(11^{-6})^5 = 11^{-30}$ shown in working **or** a correct equation with indices only (no marks for $3.914....\times 10^{-36}$)
		Working required	−34		A1	dep on M1 (as we have asked for working)
						Total 4 marks

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Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 23

23	**For 2 of** $30=5\times6$ or $30=5\times2\times3$ oe (for numerator) or $\sqrt{180}=6\sqrt{5}$ oe or $\sqrt{180}=2\times3\times\sqrt{5}$ (for denominator) or $25^{2x+7}=\left(5^2\right)^{2x+7}$ or $5^{2(2x+7)}$ oe or $\left(\sqrt{5}\right)^{4x+9}=\left(5^{\frac{1}{2}}\right)^{4x+9}$ or $5^{\frac{1}{2}(4x+9)}$ oe or $\frac{30}{\sqrt{180}}=\sqrt{5}$ oe or 5^{4x+15} as the numerator or 5^{2x+5} as the denominator		3	M1	or 5^{4x+15} as the numerator or 5^{2x+5} as the denominator or $30=5\times6$ or $30=5\times2\times3$ oe or $\sqrt{180}=6\sqrt{5}$ oe or $\sqrt{180}=2\times3\times\sqrt{5}$ or $25^{2x+7}=\left(5^2\right)^{2x+7}$ or $5^{2(2x+7)}$ oe or $\left(\sqrt{5}\right)^{4x+9}=\left(5^{\frac{1}{2}}\right)^{4x+9}$ or $5^{\frac{1}{2}(4x+9)}$ oe or
	$\frac{6\times5\times5^{4x+14}}{6\times5^{0.5}\times5^{2x+4.5}}$ oe eg $\frac{5^{4x+15}}{5^{2x+5}}$ or $\frac{\sqrt{5}^{8x+30}}{\sqrt{5}^{4x+10}}$ oe or $\frac{25^{2x+7.5}}{25^{x+2.5}}$ oe			M1	Correct expression in terms of 6 (or 2 and 3) and 5 with indices Some cancellation could have taken place **or** fully correct with $\sqrt{5}$ and powers **or** 25 and powers
	working required	5^{2x+10}		A1	dep on M2 allow $w=2x+10$
					Total 3 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 21

21	$12^2=2^4\times3^2$ **or** $2\times12^2=2^5\times3^2$ oe **or** $\frac{2\times12^2}{3^2}(=32)=2^5$		5	M1	
	$18^{4n}=(2\times3^2)^{4n}$ **or** $2^{4n}\times3^{2\times4n}$			M1	
	$3n^2-14n-5\ (=0)$			A1	
	e.g. $(3n+1)(n-5)(=0)$ $n=\frac{14\pm\sqrt{(-14)^2-(4\times3\times-5)}}{2\times3}$			M1	for solving their 3 term quadratic equation using any correct method - if factorising, allow brackets which expanded give 2 out of 3 terms correct (if using formula or completing the square allow one sign error and some simplification – allow as far as e.g. $\frac{14\pm\sqrt{196+60}}{6}$ oe)
		$-\frac{1}{3}$, 5		A1	Allow −0.33 or better for $-\frac{1}{3}$
					Total 5 marks

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Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 24

24	2^3 **and** 2^{4x} or $(2^4)^x$		5	M1	for writing 16^x **and** 8 as a power of 2 (or all as powers of 4,8 or 16)
	$n = x^2 + 4x + 3$ oe or $x^2 + 4x + 3 - n = 0$			A1	for writing n in terms of x correct expression implies first M1
	$(n=)(x+2)^2 - 2^2$ ..... oe or $(x=) -2 \pm \sqrt{n+1}$ $(x=)\dfrac{-4 \pm \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe			M1	for a correct first step in completing the square or using the quadratic formula correctly ft their 3 term quadratic
	$(x=) -2 + \sqrt{n+1}$ oe or $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe			A1	for correctly rearranging to make x the subject (must be positive square root)
	Correct answer scores full marks (unless from obvious incorrect working)	$(x=) -2 + \sqrt{n+1}$ **and** $n > 3$		A1	must be positive square root Accept $(x=)\sqrt{n+1} - 2$ oe and $3 < n$ Accept $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe **and** $n > 3$ or $3 < n$
					Total 5 marks

24 ALT	$4^{\frac{1}{2}n}$, $4^{\frac{1}{2}x^2}$, 4^{2x} **and** $4^{\frac{3}{2}}$ | $8^{\frac{1}{3}n}$, $8^{\frac{1}{3}x^2}$ **and** $8^{\frac{4}{3}x}$ | $16^{\frac{1}{4}n}$, $16^{\frac{1}{4}x^2}$ **and** $16^{\frac{3}{4}}$		5	M1	for all as powers of 4 or 8 or 16
	$n = x^2 + 4x + 3$ oe or $x^2 + 4x + 3 - n = 0$			A1	for writing n in terms of x correct expression implies first M1
	$(n=)(x+2)^2 - 2^2$ ..... oe or $(x=) -2 \pm \sqrt{n+1}$ $(x=)\dfrac{-4 \pm \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe			M1	for a correct first step in completing the square or using the quadratic formula correctly ft their 3 term quadratic
	$(x=) -2 + \sqrt{n+1}$ oe or $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe			A1	for correctly rearranging to make x the subject (must be positive square root)
	Correct answer scores full marks (unless from obvious incorrect working)	$(x=) -2 + \sqrt{n+1}$ **and** $n > 3$		A1	must be positive square root Accept $(x=)\sqrt{n+1} - 2$ oe and $3 < n$ Accept $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe **and** $n > 3$ or $3 < n$
					Total 5 marks

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Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 19

19	(a)	eg $(2^3)^2\times\sqrt[3]{(2^2)^6}$ **or** $(2^3)^2\times(4)^{\frac{6}{3}}$ **or** $4^3\times4^2$ **or** 2^6 or 2^4 seen **or** $2^6\times16$ or 64×4^2 or $8^2\times4^2$ or $8^2\times16$ or 64×16			3	M1 a correct first stage.
		$2^6\times(2^{12})^{\frac{1}{3}}$ **or** 1024 **or** 32^2 **or** 4^5 or $2^6\times2^4$				M1 dep on 1st M mark.
				2^{10}		A1 dependent on first M1 isw if 2^{10} seen but then 10 given as answer.
	(b)	$(n^{-\frac{4}{5}}=)\frac{1}{16}$ or 0.0625 oe	eg $\left(n^{-\frac{1}{5}}\right)^4=\left(\frac{1}{2}\right)^4$		4	M1 for sight of $\frac{1}{16}$ oe, even if raised to an incorrect power. **or** for algebraic approach, separating out the 4, or 5 or -1 in the power
		$(n=)\ 16^{\frac{5}{4}}$ or $0.0625^{-\frac{5}{4}}$ oe $(n=)\ 2^5$ or $\sqrt[4]{1048576}$ oe or $\frac{1}{0.0625^{\frac{5}{4}}}$ or $\left(\frac{1}{16}\right)^{-\frac{5}{4}}$	eg $(n=)\ \left(\frac{1}{2}\right)^{-5}$			M2 for a correct expression for n (M1 for one correct algebraic stage eg $n^{-\frac{1}{5}}=\frac{1}{2}$)
				32		A1
						Total 7 marks

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